

Some Functional Screens

Pain is present because the back is currently a weak link. Functional tests help to sort out why the back is a weak link and to indicate what is required to bolster the deficit. Some tests are posturally based; others assess motion control.

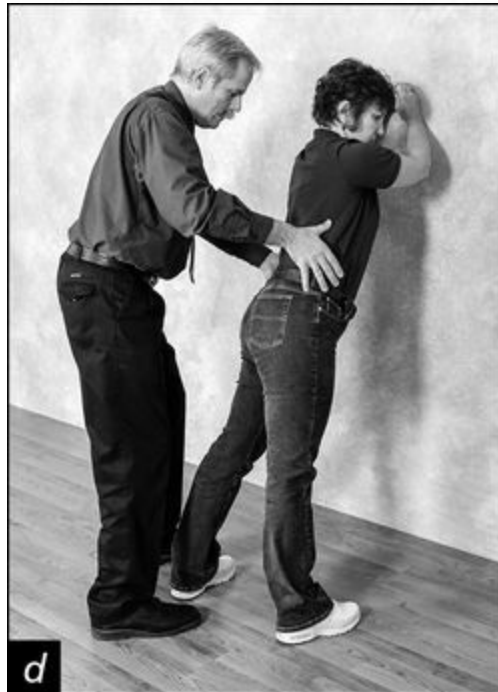
Manual Tests for Motion Control

Motion control requires all of the attributes of physical performance—motor control, strength, endurance, specific flexibility, and so on. The following screens will help you identify some of these deficits while also serving as qualifying tests prior to engagement in more progressive therapeutic exercise.

Proprioception Acuity

The ability to find a pain-free position for the spine appears to be compromised in those with chronic backs. Clearly, this ability can be trained (Preuss, Grenier, and McGill, 2005). Recall the wall plank that was part of the stop twist movement tool described in chapter 8. This drill can also be used first to identify those who are unable to accurately reposition the spine and second as a training exercise.



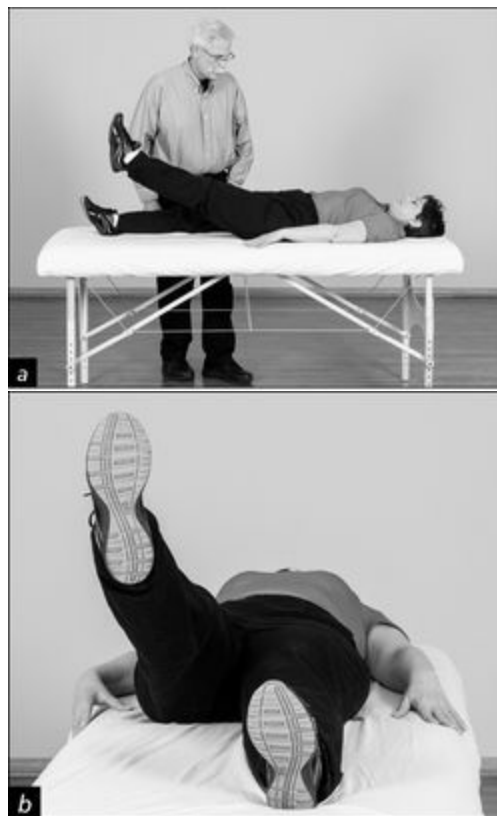


Wall planks begin with the patient's finding and grooving the sweet spot for minimal or no pain. You then move the pelvis toward the wall using a combination of hip and spine flexion–extension (a). Once the no-pain position is found, (b) rake the patient's abdominal wall with your fingers to stimulate oblique muscle activation and lock this pain-free posture into muscle memory. (c) The shoulder and latissimus may be brushed to cue the

locking initiation of the torso rotation with lastissimus dorsi. (d) Finally, groove the locked torso into the movement engram of the patient.

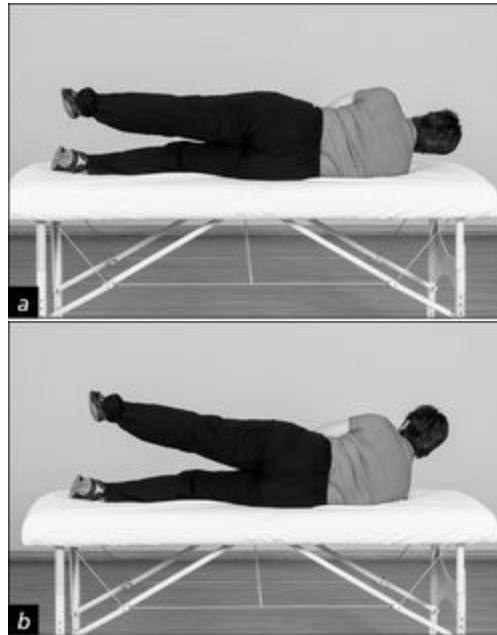
Once the pain-free posture is found, the patient spins on the balls of the feet with the rib cage locked to the pelvis so that no lumbar motion occurs (c, d). The plank position is then taken once again. If the sweet spot for pain-free lumbar posture has been lost, the exercise is repeated to groove the pain-free position into the movement and posture repertoire.

Other tests provide indicators of movement control and, by default, pain control. For example, the straight-leg raise test (SLRT) has been shown to test the ability to control the lumbar spine during a single-leg raise with the associated hip flexion (Liebenson et al., 2009) and to determine whether this is a pain trigger. Here, the patient raises a straight leg while the clinician looks for any aberrant spine twist or flexion. If the spine twists, the patient is coached to eliminate it (a, b). This allows the clinician to determine if the problem is “software” or poor understanding of the movement (or “hardware”) where sufficient strength is lacking.



In a similar way, Nelson-Wong, Flynn, and Callaghan (2009) tested the side-lying hip abduction test (a). They noted that aberrant spine movement

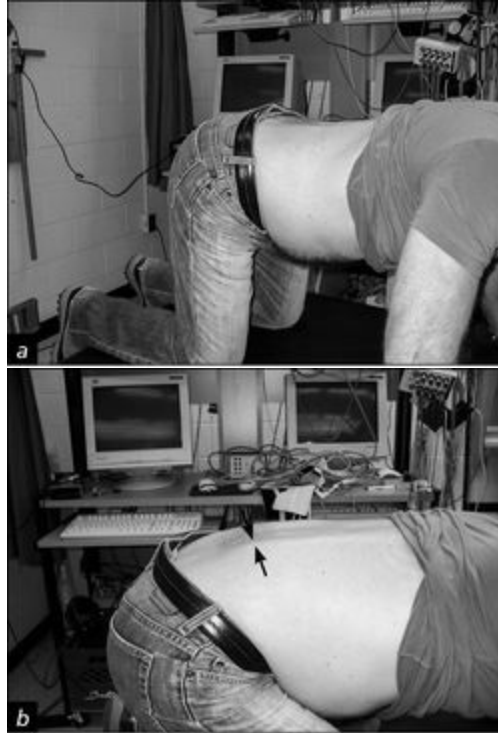
during the test predicted which asymptomatic people would develop back pain earlier during a prolonged standing task (b).



The evidence supports the notion that the ability to control spine movement during the production of hip moment reduces pain onset. Part a shows good control while part b shows poor control.

Looking for Spinal Hinges

Hinges are motion segments that bear more of their share of the motion. In the sagittal plane they are seen during sitting down or standing flexion. A common location is the thoracolumbar junction. In athletic populations, a successful strategy is often to buttress this lack of stiffness with latissimus dorsi activation through shoulder depression with the rib cage lifted (see McGill, 2006). Lumbar hinges are usually better buttressed with an abdominal brace (within the compression tolerance of the spine and at the sufficient level). Sometimes torsional hinges occur. These can be addressed by learning to avoid painful twisting or by buttressing with an appropriate muscular brace (or both). Over time, the natural degenerative cascade will work in favor of the patient to stiffen the painful segment. In this way the arthritic or degenerative process acts to stabilize the painful segment (or hinge). However, the clinical objective is to restore pain-free function in the meantime.



Photos from Stuart McGill

The test begins with the hips over the knees and the shoulders over the hands (a). The patient rocks the buttocks back to the heels, trying to maintain a neutral spine. A flexion hinge (arrow) at L2-L3 is the site of pain in this flexion-intolerant former athlete. General abdominal bracing removed his pain together with spine–pelvis position awareness training.

The Stiff Spine

Not all spines are painful because of instability. Rather, some are stiffened with muscular contraction and locked into painful postures. The chronic muscular lock, however, is often secondary to tissue damage and local instability. However, the choice of coping strategy is what causes the chronic symptoms. The muscular reaction is simply inappropriate: It is probably far too intense and is more than is needed to be sufficient. Thus the brace is tuned to the task—just enough to control pain. As noted earlier, stiffness is often regional, and various clinical approaches can be employed, ranging from passive approaches to active learned motion patterns that involve less muscular stiffening. Techniques for spine mobility, when appropriate, are shown later.

Control of Torsional Motion

Other forms of aberrant motion may indicate poor lumbar control, which is in contrast with the local segmental instabilities shown previously. For example, a rotational disconnect of the rib cage and pelvis during the push-up test described here can indicate that more work is required for enhancing lumbar control.

Torsional Control Tests

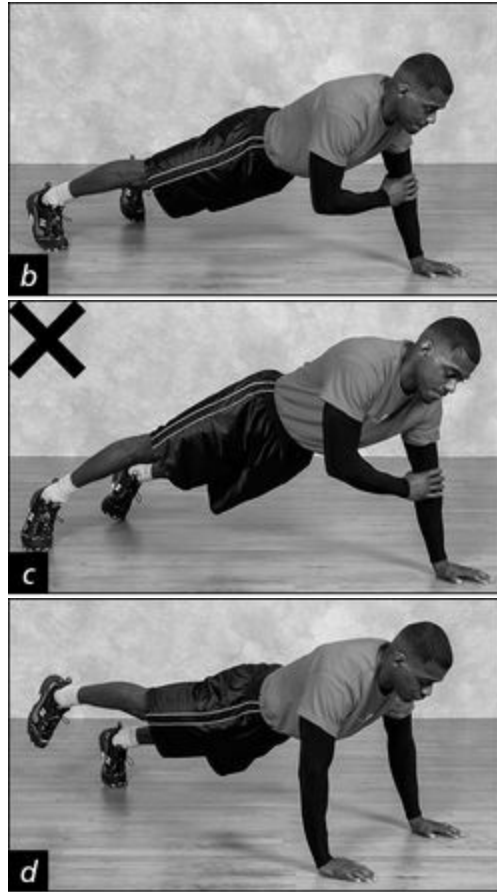
The stop twist movement tool can be considered to provide an initial impression of torsional control. **Begin with the wall plank test and assess standing twist competency.** The more advanced torsional tests described here indicate the presence or absence of the control necessary before engaging in more demanding torsional training.

Push-Up Tests

After the patient has assumed a correct push-up position (a), have him lift one hand in a controlled motion and place it over the opposite elbow (b). This is good form and shows torsional control. Poor control is shown in (c), contraindicating further torsional training. The pelvis and rib cage should remain level if torsional control is achieved.

The one-legged hip extension from the push-up position provides a similar indicator of control (or lack thereof). A pelvis locked to the rib cage while raising one leg with hip extension also indicates lumbar torsional control (d). Hiking the pelvis indicates poor lumbar torsional control.





Back Bridge Test

The back bridge with a leg extended also indicates torsional control. Although this was originally proposed as a test for pelvic stability (Mens et al., 1999), we have found that it reveals lumbar torsional control, stiffness, and stability (Liebenson et al., 2009).



Have the patient get into the back bridge position; then ask the patient to extend one leg slowly. The patient should be able to control this motion with no twisting or flexion in the lumbar region.

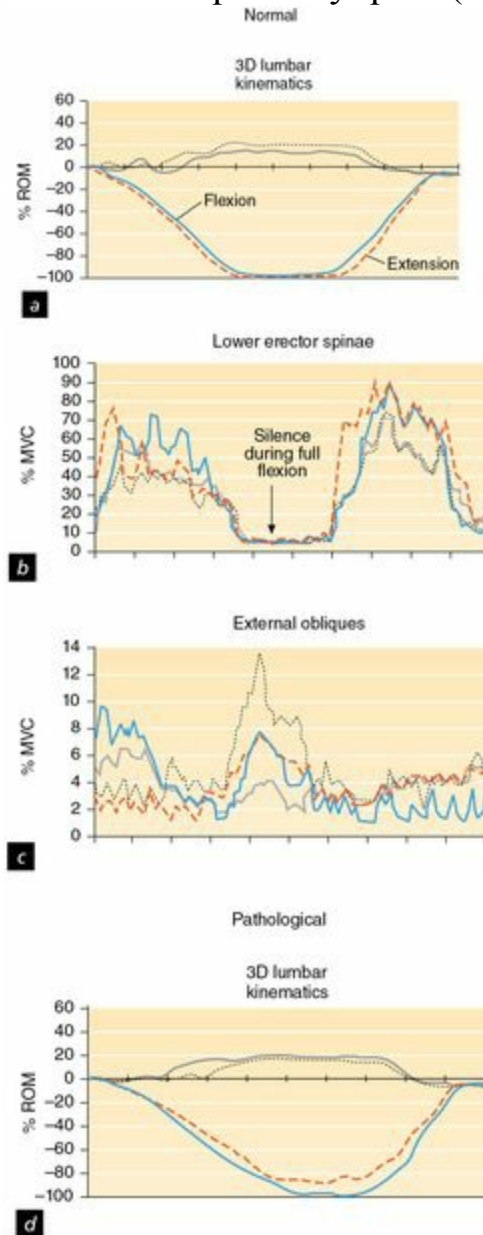
Testing for Aberrant Gross Lumbar Motion

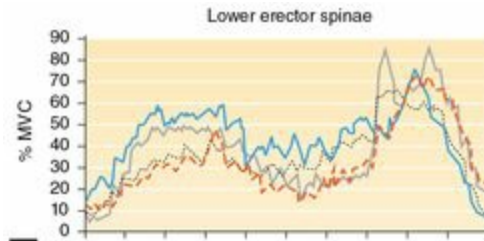
Local spine instabilities are detected in several ways. The more skilled clinicians simply notice them. They appear as small deviations in the three-dimensional motion profiles. For example, the spine may clunk during standing flexion when the lumbar spine bends through 14° . This is repeatable once the refractory period has passed. The refractory period is not a set time but a condition in which the unstable joint is able to shift back to a neutral position so that it is able to clunk again. The clunks are almost always manifested in a shear mode. Sometimes they are audible with the naked ear or with a stethoscope. Other times they are a simple and very subtle shift in motion as the spine is moving in a specific way, but they cause pain at that instant. Correction exercises involve avoiding the motion that causes the pain, buttressing the instability with an appropriate bracing pattern, or both. A skilled clinician will see these instabilities through unaided observation; however, I have also found technology helpful, particularly when I have to document these cases for medical and legal reports. The following are a few examples.

Aberrant Gross Lumbar Motion and Electromyographic Evidence

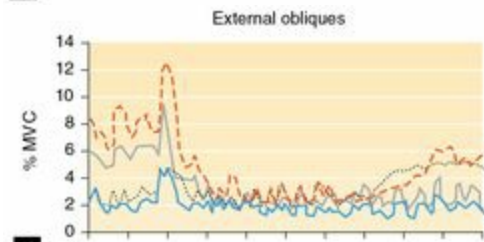
The usual practice of assessment is to test for ROM about the three primary axes of motion by determining the degrees of motion possible from neutral. For example, a flexion ROM may be determined to be 45° from neutral (usually using standing posture as a reference base). As previously pointed out, this number is of little use as an indicator of pathology or for guiding therapeutic exercise decisions. Other well-known pathology markers include the inability to achieve myoelectric silence in the extensors at full flexion (and that full flexion is not reached—see [figure 9.2](#)). However, pathology may manifest within moderate ranges of motion, and simultaneously in axes other than the primary plane of motion. The following example illustrates this point. As one bends forward in the sagittal plane, a small amount of motion may occur in the lateral bend or axial twist axis, and it may be repeatable every time the spine passes a certain flexion position. This indicates an instability catch, or clunk. These are seen in the examples shown in [figures 9.3](#) and [9.4](#).

Figure 9.2 Flexion response test. The normal person flexes forward to full flexion (a), shows silence in the erector spinae muscles (b) and actually pulls with the obliques (c). The person with pathological pain does not reach full flexion to transfer load from muscle to passive tissues (d), does not shut off the erector spinae (e), and the obliques stay quiet (about 3% MVC) (f).





e



f

Figure 9.3 This patient had a normal ROM but demonstrated an instability catch. During a lateral bend test (from neutral 0 to 23°), every time the spine bent laterally passing 17° (event A) (b), a small flexion hitch of a degree and a half occurred in the flexion axis (event B) (a). There was no deviation in the twist axis (c). The bottom curve represents the first derivative of the flexion curve, which we use to better locate instability. The spike (event C) (d) clearly indicates the point at which the instability occurred in the flexion axis. In this patient, a muscle activation pattern was identified that removed the painful clunk, or catch.

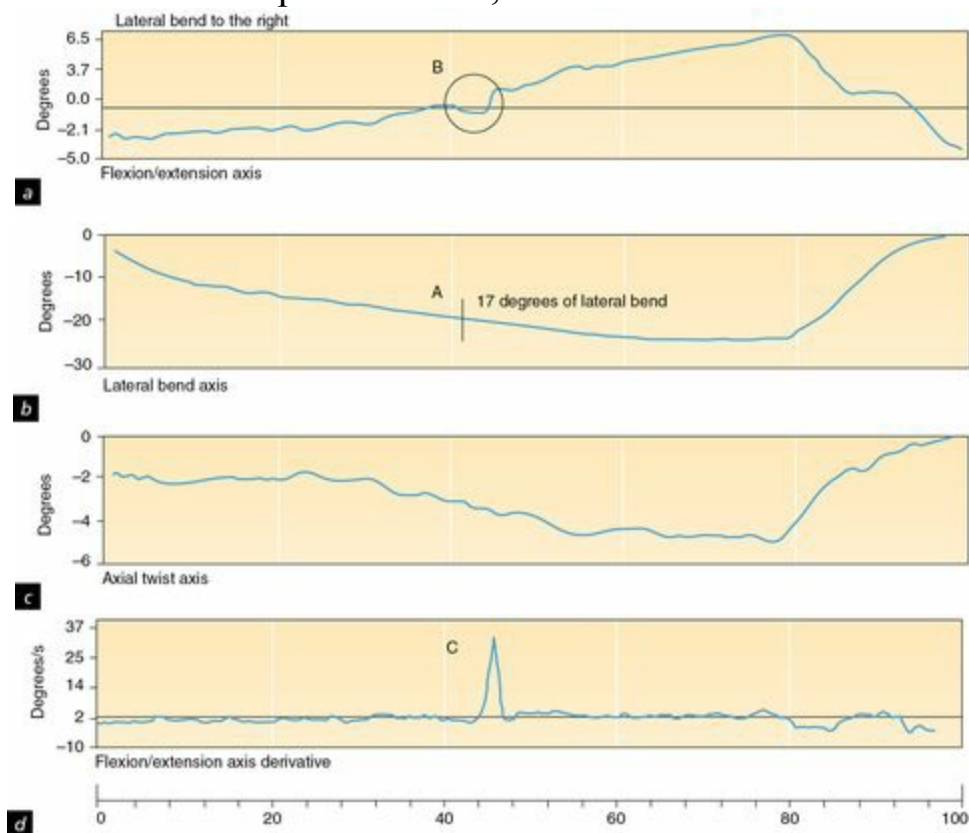
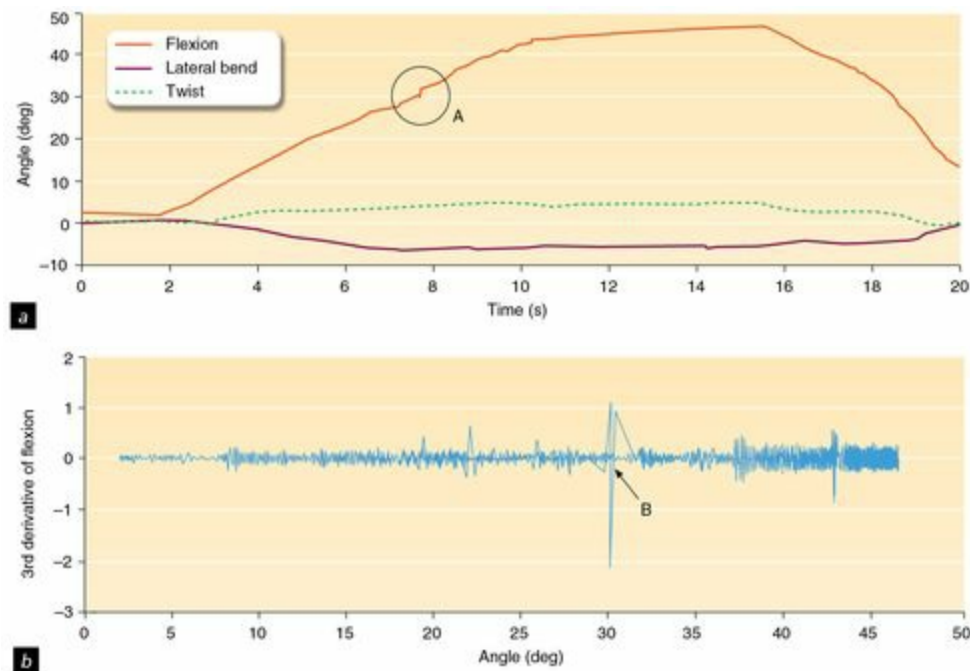


Figure 9.4 In this patient an instability catch, or clunk, occurred in the primary axis of motion (flexion). As he flexed, the clunk was seen in the flexion axis at 30° (event A) (a). This occurred in every trial of forward flexion. The instability is evident (event B) in the third derivative of the flexion axis motion (b). The rehabilitation objective was to stabilize the spine to eliminate the clunk. This was not fully successful over the short term.

Symptoms resolved only when the patient developed spine position awareness and was able to avoid flexion to 30°. Specifically, in this case, the instruction was a mild abdominal brace.



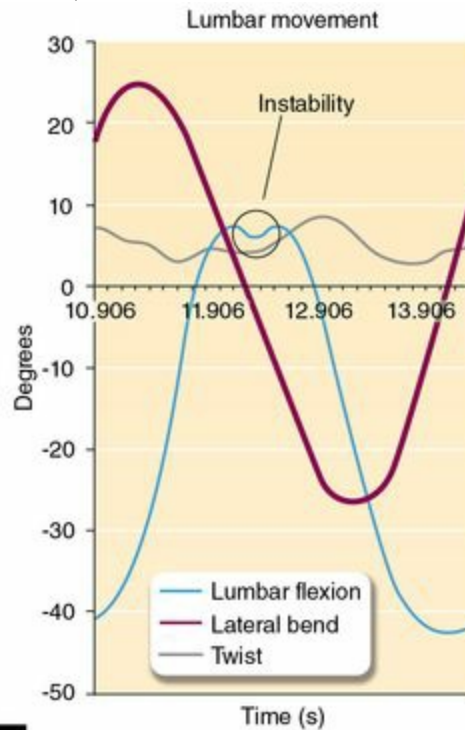
The first patient ([figure 9.3](#)) had a normal ROM and by traditional tests would have been classified as normal. Yet during a lateral bend test (from neutral 0 to 23°), every time the spine bent laterally passing 17°, a small flexion hitch occurred in the flexion axis of a degree and a half. This was repeatable. The point is that the motion pathology was not in the lateral bend axis but was seen in the flexion axis. These very subtle aberrant catches in the “off” axes are often missed during examination. The more skilled clinicians see these or feel them with their hands.

In the second example ([figure 9.4](#)), a clunk, or catch, was observed in the flexion axis as the patient flexed forward and passed 30°—this was repeatable and observed in every flexion attempt. Interestingly, this

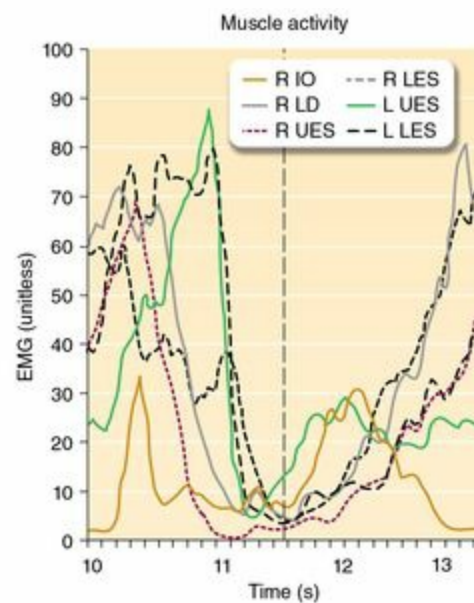
instability catch, as well as the associated pain, was cured with some muscle patterning and stiffening in a very short period of time. Specifically, in this case, the instruction was a mild abdominal brace. Sometimes patients report pain as the catch occurs, and sometimes there is evidence of muscle activation bursts; but many patients simply move through the region. We use an electromagnetic instrument that tracks three-dimensional spine motion and graphically displays the results. Sometimes the catches are so subtle that it is far more illuminating to take the first derivative of the axis with the catch. Patients have a much higher incidence of these catches (Peach and McGill, 1998) than people without back issues, which indicates instability. People without symptoms who demonstrate these catches may be at elevated risk for becoming patients according to the aberrant motion model of Richardson and colleagues (1999).

A final example is very striking. One patient arrived after seeing many specialists who told him that his problems were in his head. He had intransigent sciatica and could not stand the weight of a bedsheet on his foot. The specialists had treated his foot! I asked him what caused his pain, and he started to wind his upper torso around. As he approached upright standing, I heard a grind and pop from his low back and he yelled out in pain. Clearly, he had a massive spinal instability with a trapped lumbar nerve root (see [figures 9.5](#) and [9.6](#)). Upon looking at his motion and muscle electromyogram (EMG) data, I could see that his spine became unstable when the muscles shut down as he returned to an upright posture. His nerves were highly sensitized, accounting for his seemingly bizarre (to others—not to us) behavior. We showed him some muscle-bracing patterns, which removed the audible clunk within about 4 days. After rapid progress, he was pain free in about 3 months. Bracing took away the daily irritation to the sensitized tissues. The bracing then also allowed his tissues to desensitize. We have done this time and time again with patients who have had no success with other approaches—the approach can be very successful with these unstable patients. We have seen many patients who were suicidal with their pain and who were told the problem was in their heads. It is tragic.

Figure 9.5 (a) The three-dimensional motion of the lumbar spine is shown with the instability catch occurring as the patient returned to an upright posture. (b) Observing the EMG records revealed that this was the instance in time when muscle activation (and stiffness and stability) was at a minimum.

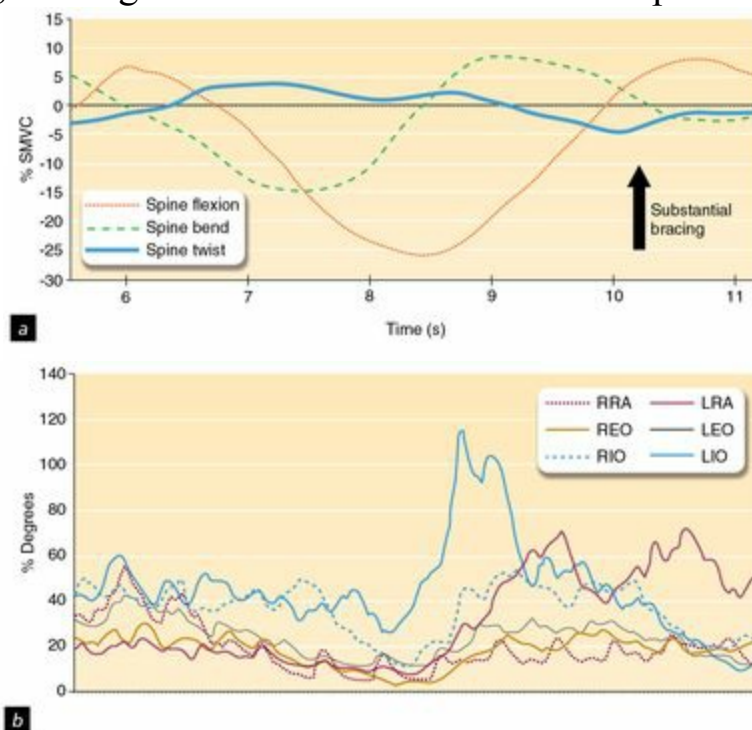


a



b

Figure 9.6 The same patient as figure 9.5 was tested three months later. Here, they performed the same movement and (a) no clunk is evident as they (b) learned to brace, causing stiffness sufficient to eliminate potential instability.



Testing for Aberrant Motor Patterns During Challenged Breathing

The test for aberrant motor patterns during heavy breathing is rather sophisticated, but it is probably the only way to identify a person's ability to maintain spine stability during tasks that require higher physiological work rates. Candidates for this type of test include athletes and occupational workers such as construction workers and warehouse employees. We explored the link between breathing and spine stability in chapter 8. Challenged breathing requires heavy involvement of the abdominal muscles. Because we monitor the muscles with EMG electrodes and because muscle fatigue changes the EMG signal, we avoid creating heavy lung ventilation with exercise for this test. (Note that this consideration is only for testing; we intentionally increase the physiological work rate for exercises in chapter 5.)

Challenged Breathing Test

During this test the person breathes a 10% CO₂ gas mixture to elevate the lung ventilation with only a few breaths, but the O₂ is adjusted so that the light-headedness that accompanies hyperventilation does not occur. Then a weight, usually about 15 kg (33 lb) for the average man, is placed in the hands while the torso is flexed forward about the hips to approximately 30°. This test reveals those who have aberrant motor patterns in the abdominal muscles during challenged breathing that compromise lumbar stability.



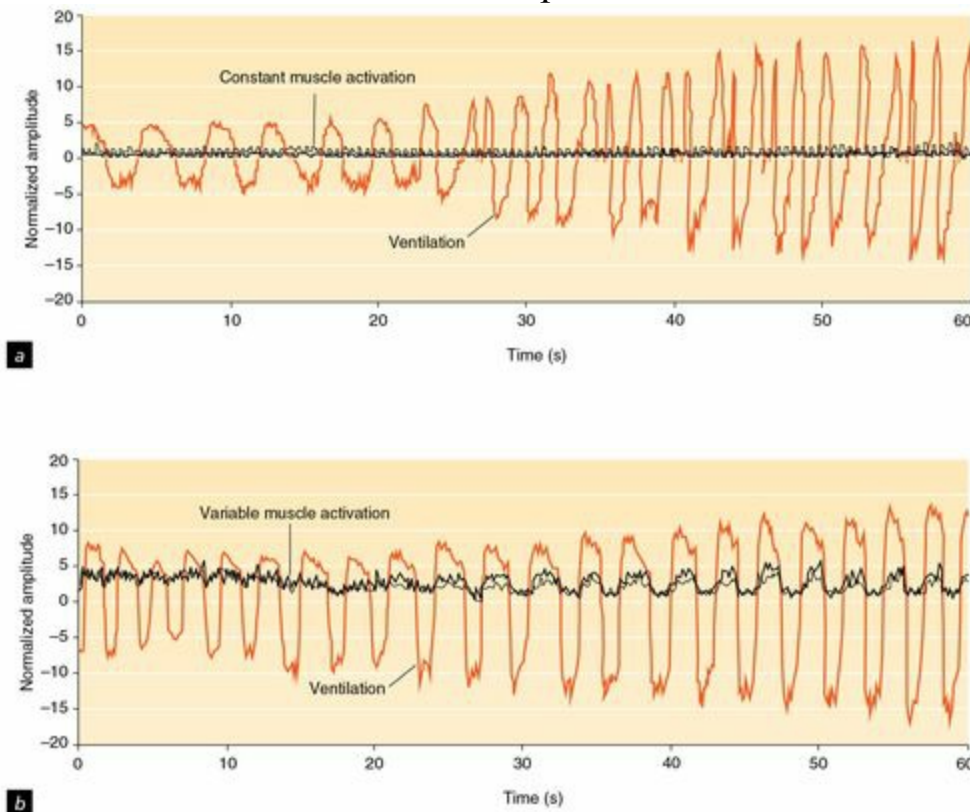
Photos from Stuart McGill

Good spine stabilization patterns observed in the muscle EMG signal obtained during the challenged breathing test include a constant muscle cocontraction ensuring spine stability. [Figure 9.7](#), a and b, shows stable and unstable patterns. Our computerized analysis package measures spine stability using the methods explained in chapter 6. Many clinicians will not have this capability but should still look for constant levels of abdominal wall activation. Patients who are poor at stabilization in these situations will demonstrate short, temporary reductions in activation or even loss of activation. These mark the critical instances at which spine instability can occur, and classify the person as a candidate for spine stabilization during high physiological work rates. These patients must learn to contract the spine

stabilization muscles independently of challenged breathing to ensure spine stability in conditions of heavy work.

Figure 9.7 Good spine stabilization patterns during the challenged breathing test include a constant muscle cocontraction, which ensures spine stability.

Both (a) stable with constant muscle activation and (b) unstable variable abdominal muscle activation patterns are shown.



Many other provocation tests and functional screens are suited for more athletic populations. Some can determine the starting load for serious resistance training, whereas others indicate basic movement flaws that must be corrected prior to rigorous training such as with plyometrics. A few of these are shown in *Ultimate Back Fitness and Performance* (McGill, 2014).

Testing Muscle Endurance

Several tests that can help you assess your patients and plan the most appropriate exercise are described in this section. You are the final judge as to which tests are relevant for specific patients.

Muscular Endurance Tests

Although Biering-Sorensen (1984) showed that decreased torso extensor endurance predicts those who are at greater risk of back troubles, recent work has suggested that the balance of endurance among the torso flexors, extensors, and lateral musculature better discriminates those who have had back troubles from those who have not. Because these three muscle groups are involved in spine stability during virtually any task, the endurance should be measured in all three. Simple tests that isolate these groups of muscles are difficult to find; I chose the following tests because each was shown to have high reliability coefficients (at least .98 or higher) when repeated over 5 consecutive days (McGill, Childs, and Liebenson, 1999).

Lateral Musculature Test

The lateral musculature is tested with the person lying in the full side bridge position (the test is also known as the side bridge or side plank). Legs are extended, and the top foot is placed in front of the lower foot for support. Subjects support themselves on one elbow and on their feet while lifting their hips off the floor to create a straight line over their body length. The uninvolved arm is held across the chest with the hand placed on the opposite shoulder. Failure occurs when the person loses the straight-back posture and the hip returns to the floor.



Photos from Stuart McGill

Flexor Endurance Test

Testing endurance of the flexors (rectus) begins with the person in a sit-up posture with the back resting against a jig angled at 55° from the floor (a). Both knees and hips are flexed to 90° ; the arms are folded across the chest with the hands placed on the opposite shoulder; and the toes are secured under toe straps. To begin, the jig is pulled back 10 cm (4 in.) and the person holds the isometric posture as long as possible (b). Failure is determined to occur when any part of the person's back touches the jig.



Photos from Stuart McGill

What Is the Best Test?

Concerns about using speed sit-ups as a test of core fitness in the mandatory annual fitness test in the U.S. military have been discussed previously. Recognizing the link between repeated sit-ups and the incidence of low back injury, Peterson (2013) evaluated alternate measures and suggested the front plank. The plank has the benefit of not needing any equipment, which is important for those deployed on ships and aircraft. The downside is that shoulder disorders may disguise true core endurance.

McGill and colleagues (2010) compared endurance data with the V-sit flexor endurance test and the front plank in a population of 620 firefighters for the first data set and 181 university students for the second. Although

flexor endurance in firefighters peaked in their 40s, extensor endurance peaked in their 20s. In the study of university students, the plank scores were relatively higher than the V-sit scores in males but relatively lower than the V-sit scores in the females. A Pearson correlation test between the paired plank and V-sit scores of each subject rendered a coefficient of $r = .34$. This means that the [performance](#) on one flexor test predicted only 11% of the score in the other. An ANOVA comparison of scores based on subjects' history of having had shoulder or back troubles showed no significant link between V-sit and plank scores. The plank scores are not well correlated with the V-sit scores, suggesting that the two measure different variables. Because more data exist for the V-sit, measurement of flexor endurance in occupational settings using this test probably forms a stronger link to back injury.

Back Extensor Test

The back extensors are tested in the Biering-Sorensen position, with the upper body cantilevered out over the end of a test bench and with the pelvis, knees, and hips secured. The upper limbs are held across the chest with the hands resting on the opposite shoulders. Failure occurs when the upper body drops from the horizontal position.



Photos from Stuart McGill

Normative Data

Normative absolute endurance times collected from young healthy people are listed in [table 9.1](#). Note that women have greater endurance than men in the extensors. Further, the flexor endurance time (as a proportion of the extensor endurance time) for healthy young men is about .84, whereas other tests have yielded lower ratios for healthy older men. McGill and colleagues

(2003) showed that the relationship of endurance among the anterior, lateral, and posterior musculature is upset once back troubles begin (see [table 9.2](#)); that upset in the relationship remains long after the symptoms have resolved. Typically, extensor endurance is diminished relative to both the flexors and the lateral musculature in those with lingering troubles. A final note regarding patient endurance testing: There is no need to exacerbate the pain in patients with endurance testing, because their daily variation in pain will invalidate their results. Wait until they are pain free. In addition, resist the urge to “train the tests.” Rather, build endurance with the reverse pyramid approach introduced in chapter 8 and with repeated sets of shorter-duration holds.

Table 9.1 Mean Endurance Times (sec) and Ratios Normalized to the Extensor Endurance Test Score

Task	MEN			WOMEN			ALL		
	Mean	SD	Ratio	Mean	SD	Ratio	Mean	SD	Ratio
Extension	161	61	1.0	185	60	1.0	173	62	1.0
Flexion	136	66	0.84	134	81	0.72	134	76	0.77
RSB	95	32	0.59	75	32	0.40	83	33	0.48
LSB	99	37	0.61	78	32	0.42	86	36	0.50
Flexion/extension ratio	0.84			0.72			0.77		
RSB/LSB ratio	0.96			0.96			0.96		
RSB/extension ratio	0.58			0.40			0.48		
LSB/extension ratio	0.61			0.42			0.50		

Mean age of university students = 21 (men: $n = 92$; women: $n = 137$).

The ratios are determined for each person and then averaged. They are not derived simply by dividing the average scores of the group.

RSB = right side bridge; LSB = left side bridge.

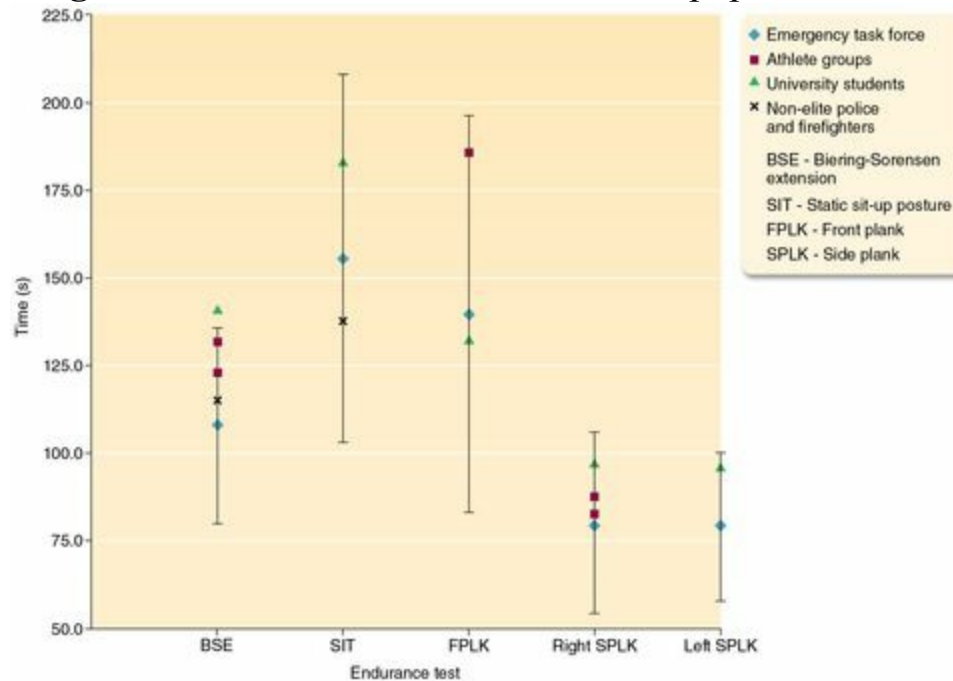
Table 9.2 Mean Endurance Times Comparing Normal Workers With Those Who Have Had Back Disorders

Task	NO BACK TROUBLES			HISTORY OF DISABLING BACK TROUBLES		
	Mean	SD	Ratio	Mean	SD	Ratio
Extension	103	35	1.0	90	49	1.0
Flexion*	66	23	0.64	84	45	0.93
RSB	54	21	0.52	58	23	0.64
LSB	54	22	0.52	65	27	0.72
Flexion/extension ratio*	0.71	0.26		1.15	0.66	
RSB/LSB ratio*	1.05	0.32		0.93	0.22	
RSB/extension ratio*	0.57	0.29		0.97	1.20	
LSB/extension ratio*	0.58	0.28		1.03	1.16	

Men, mean age 34 (never had back troubles: $n = 24$; lost work due to low back disorders: $n = 26$) from the same workplace. Variables that are significantly different between those with a history and those who have never had troubles have an asterisk. Note that all men were asymptomatic at the time of testing; these are long-lingering deficits subsequent to back troubles.

Endurance data from other populations give context to the links to back injury (see [figure 9.8](#)). Firefighters and police officers were followed over a period of 3 and 5 years, respectively, with particular focus on the links between certain fitness variables and the risk of developing back pain. Interestingly, in these professions, the fitter members had a higher risk. It appears that the injuries were more likely to occur in the station fitness training rooms (Frost et al., in press). This speaks volumes about the inappropriate training methods currently popular in these populations. Thus, the complexity of the interactions among exposure, movement competency, training, fitness, and injury in these worker groups may occlude their true relationships.

Figure 9.8 Endurance data from various populations.



Interpreting Endurance Scores

Interpreting absolute endurance is probably secondary to interpreting the relationship among the three muscle groups (flexors, lateral, and extensors). Normally, the following discrepancies suggest unbalanced endurance (note that the last ratio is a strength-to-endurance ratio).

Right-side bridge/left-side bridge endurance > 0.05

Flexion–extension endurance > 1.0

Side bridge (either side)–extension endurance > 0.75

Extensor strength (Nm)–extensor endurance (sec) > 4.0

Probably the most important ratio is the strength-to-endurance score.

Many strong men and women suffer repeated acute episodes of back pain. Most report the pain-inducing episode as resulting from a break in form. We interpret these observations to mean that endurance helps to preserve good spine-sparing form when performing tasks requiring some strength. Strength should not get ahead of the ability to control strength and retain good posture and form.

Other observations linking perturbed endurance with back disorders include those of Nelson-Wong and Callaghan (2014), who noted that those with poorer side bridge endurance scores were the first to develop pain when standing. Observing a series of case studies of professional hockey ice players, of the 5 who suffered sportsman's hernias, all had side bridge scores below 70 seconds at the beginning of the season. We are now using 70 seconds as a minimum for all players.

Several scientists have applied these scores to special populations such as elite golfers. It appears that golfers in this group have higher lateral endurance than normal. Such special groups do not conform to the average scores and rules for interpretation.

Integrating Evidence From Medical Images With Assessment Findings

There appears to be an increasing trend for physicians to order medical images for those presenting with back pain. Although this may be warranted in the presence of a red flag condition, it is rarely necessary. Some clinics advertise that if a patient sends images, they will determine whether their surgery will help them. In my view this is simply a business venture for profit. The best clinicians determine the pain causes and triggers from provocation testing. They also create the advantage of immediately determining what can take the patient's pain away. Further, the picture often bears little reflection of the amount of pain for reasons covered in chapters 1 and 2.

The next issue has to do with the use of the information contained in a radiologist's report. Offhand comments such as No tumor is present or The conus terminates at a normal level reveal that the radiologist has not been directed on what she should be looking for. The requisition should include details about the alleged or candidate injury or pain mechanism. This would

help the radiologist optimize the patient setup (e.g., changing the load or posture to enhance the chance of revealing the pathology). Remember that the radiologist without such direction has no impression of the patient, the severity of the symptoms, or the postures that decrease or increase the pain. I see case after case in which the radiology was simply bogus. For example, a patient with a dynamic disc bulge that grows with flexion postures should sit flexed for a few minutes prior to the image acquisition. Failure to do this results in a bona fide patient being told he has no problem. We see patients with massive bony fracture only to receive a report of normal findings because the radiologist did not know how to position the patient appropriately (McGill and Yingling, 1999). The same is true for spine instability. So much more could be reported to the referring clinician in terms of the nature of the instability if the radiologists had some direction about acquiring the most appropriate image coupled with training to interpret the finding in a way to guide clinical treatment. The optimal situation is the clinician and the radiologist working together to discuss the case.

Obviously, images are very helpful in certain situations. I have found a few tumors over my career that were missed by the radiologist. Syndromes such as Scheuermann's disease, for example, can be confirmed radiologically. Yet even with Scheuermann's, there appears to be a constellation of features that would alert an astute clinician. For example, all of the patients with Scheuermann's whom I have seen have been lanky males with slouched postures (I am not sure what is the cause or consequence here).

The radiological diagnosis of degenerative disc disease has been discussed at several junctures in this book. This is not a disease. A thinned disc at one level should be followed up to examine the integrity of the end plate; most often this reveals damage, allowing the radiologist to report a useful directive rather than a red herring.

I also believe that it is well overdue for radiologists to become better schooled in the causes of the features they report. For example, Modic changes along the margins of vertebrae, in the presence of radiological instability, indicate a bone bruise. Clinically, this results in load intolerance for substantial periods of time. The most insightful monograph linking radiological findings with clinical features is Morgan (2013). A report of the muscle density and neurologically dead zones observed within muscle would also be helpful, yet these are not considered standard components of a report. Sadly, I never look at the radiology report anymore; I insist on seeing the

original pictures.

A Final Note

The process of patient assessment is never ending. I find that my assessments take longer as I become better at probing the pain mechanism and the mechanisms responsible for aberrant movement patterns. But the effort is worth it. At the end of my assessments, the pain mechanism is almost always known, and the exact pain triggers are uncovered. (If you are unable to converge on a mechanical diagnosis, there is a good chance there is a more sinister cause of the pain.) Then I create the plan to help the patient avoid the pain triggers followed by progressions of exercise that address the precise clinical targets.